

Research Article

Alexandr Udeneev and Oleg Bakhteev

Extracting Inductive Bias Using Generative Models

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Abstract: Inductive biases shape how neural networks generalize from limited data, yet encoding them in network connectivity often relies on domain expertise. This paper presents a generative approach that draws on related tasks to discover shared structural constraints. For each task, a generative model approximates a family of hypothesis spaces that admit near-optimal solutions. Intersecting these task-specific families narrows the search to structures compatible with all tasks, automating the discovery of a shared inductive bias (want lossless). Experiments on synthetic tasks demonstrate improved data efficiency and generalization to unseen tasks, supporting the hypothesis that the recovered constraints capture transferable structure.

Keywords: inductive bias, generative model

1 Introduction

Choosing an appropriate inductive bias is crucial for generalization [1]. By constraining the hypothesis space to reflect the underlying task structure, such a bias enables effective learning from limited data. In neural networks, examples of structural inductive biases include local connectivity and spatial weight sharing in convolutional architectures [3], and recurrent state updates with parameters shared across time in recurrent architectures [2]. Suitable parameter-sharing patterns also enforce equivariance to task-relevant transformations [4]. Building on these principles, recent work has explored learning symmetry-inducing parameter-sharing patterns directly from data [5, 6].

Selecting suitable structural biases by hand requires domain expertise, particularly when the relevant properties of a new task are not known in advance [5]. This motivates learning such biases from data. For example, Zhou et al. [6] represent layer weights as the product of a transformation matrix shared across tasks and a vector of task-specific parameters. The shared matrix is meta-learned across tasks to capture transferable parameter-sharing patterns, including those that induce equivariance. For larger layers, the authors use a Kronecker-factorized approximation to reduce memory and computational costs, restricting the search to a structured family of reparameterizations. More generally, a single task may admit multiple near-optimal structures, and its data alone need not identify which constraints will transfer to other tasks. The broader challenge is therefore to identify structural constraints that support learning across an environment of related tasks and generalize to previously unseen tasks from that environment [1].

2 Related Works

TODO

3 Preliminary

3.1 Tasks and hypothesis spaces

We adopt the formulation of inductive bias learning introduced by Baxter [1]. Let $\mathcal{X}$ and $\mathcal{Y}$ denote the input and output spaces, and let $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, \infty)$ be a loss function. A supervised learning task is represented by a probability distribution P on $\mathcal{X} \times \mathcal{Y}$. A hypothesis is a function $h : \mathcal{X} \rightarrow \mathcal{Y}$, whose population risk is

$$R_P(h) := \mathbb{E}_{(x,y) \sim P} [\ell(h(x), y)]. \quad (1)$$

We assume that the expectations appearing below are well defined and finite.

For a single task, a task learner receives a sample $S = \{(x_j, y_j)\}_{j=1}^m$ drawn independently from P . Its empirical risk is

$$\hat{R}_S(h) := \frac{1}{m} \sum_{j=1}^m \ell(h(x_j), y_j). \quad (2)$$

A task learner $A_{\mathcal{H}}$ maps this sample to a hypothesis $A_{\mathcal{H}}(S) \in \mathcal{H}$ within a given hypothesis space $\mathcal{H}$. The choice of $\mathcal{H}$ encodes an inductive bias: restricting the available functions can facilitate generalization from limited data, provided that the space retains sufficiently good solutions.

3.2 Task environment and bias learner

Let Q be a probability distribution over a collection $\mathcal{P}$ of task distributions. Source tasks $P_1, \dots, P_n$ are drawn independently from Q , and each supplies a sample S_i of independent observations. Given a family of hypothesis spaces $\mathbb{H}$, a bias learner B uses these samples to select a shared hypothesis space:

$$\hat{\mathcal{H}} := B(S_1, \dots, S_n) \in \mathbb{H}, \quad (3)$$

Task learners use the selected space to learn their own hypotheses on new tasks drawn independently from Q . Following Baxter, the suitability of a space for this environment is measured by

$$R_Q(\mathcal{H}) := \mathbb{E}_{P \sim Q} \left[\inf_{h \in \mathcal{H}} R_P(h) \right]. \quad (4)$$

The task-specific infimum allows different hypotheses to solve different tasks within the same shared space.

3.3 Connectivity as a structural inductive bias

In our setting, hypothesis spaces are indexed by connectivity masks. Fix a network architecture with E candidate connections in the layer whose structure is to be learned. A binary mask

$$M \in \mathcal{M} \subseteq \{0, 1\}^E \quad (5)$$

specifies which connections are available. The admissible set $\mathcal{M}$ may impose a fixed connection budget or additional connectivity constraints.

Let $f_{w,M}$ denote a predictor with mask M and numerical parameters w . Masked connections are set to zero, while w includes the active weights and the remaining trainable parameters. Each mask defines a hypothesis space

$$\mathcal{H}_M := \{f_{w,M} : w \in \Theta_M\}, \quad \mathbb{H}_{\mathcal{M}} := \{\mathcal{H}_M : M \in \mathcal{M}\}, \quad (6)$$

where Θ_M is the admissible parameter set.

Learning a structural bias therefore amounts to selecting a mask whose induced space is suitable for the task environment. The corresponding population objective is

$$\inf_{M \in \mathcal{M}} R_Q(\mathcal{H}_M) = \inf_{M \in \mathcal{M}} \mathbb{E}_{P \sim Q} \left[\inf_{w \in \Theta_M} R_P(f_{w,M}) \right]. \quad (7)$$

In this setting, the bias learner B selects a mask $\widehat{M}$ and thereby returns the space $\widehat{\mathcal{H}} = \mathcal{H}_{\widehat{M}}$. The selected mask is fixed across tasks, while a task learner trains numerical parameters from scratch for each new task. Connectivity constrains the support of the weights; equality constraints between active weights require an additional parameter-sharing mechanism.

3.4 Learning new tasks from finite samples

The environment risk measures the best solutions available within a hypothesis space. To describe learning from limited data, we also specify how a task learner uses that space.

For a fixed mask M , let A_M denote the task learner operating within $\mathcal{H}_M$. It returns a predictor $A_M(S; \xi) \in \mathcal{H}_M$ using sample S and randomness ξ , including fresh parameter initialization. Its expected risk after observing m labeled examples is

$$\mathcal{R}_m(M) := \mathbb{E}_{P \sim Q} \mathbb{E}_{S \sim P^m, \xi} [R_P(A_M(S; \xi))]. \quad (8)$$

For a mask $\widehat{M}$ selected by the bias learner B , this quantity is evaluated conditional on the source data and the selected mask. The new task and its sample are independent of this selection.

Our objective is to learn a fixed structure that improves $\mathcal{R}_m$ at limited sample sizes, or reaches a prescribed risk level with fewer examples than comparison structures under a common training protocol. A low value of $R_Q(\mathcal{H}_M)$ establishes that good solutions are available. Whether the training procedure can obtain them efficiently from finite samples is a separate question.

4 Method

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5 Experiments

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6 Conclusion

TODO

References

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